

Product Requirements Document (PRD)

CarbonWise

Version: 1.0

Status: Draft

1. Executive Summary

CarbonWise is an intelligent sustainability assistant that helps individuals understand, track, and reduce their carbon footprint through personalized recommendations, behavioral insights, and measurable sustainability goals.

Unlike traditional carbon calculators that simply estimate emissions, CarbonWise continuously analyzes user behavior, identifies emission hotspots, and recommends simple, actionable changes that maximize environmental impact while minimizing user effort.

The platform acts as a personal carbon coach that transforms sustainability data into practical decisions.

2. Problem Statement

Individuals contribute to carbon emissions through transportation, food consumption, energy usage, and purchasing behavior, but most people:

- Do not understand their personal carbon footprint
- Cannot identify major emission sources
- Lack personalized guidance
- Struggle to maintain sustainable habits
- Cannot measure progress effectively

As a result, sustainability awareness rarely translates into long-term behavior change.

3. Product Vision

Enable every individual to make environmentally responsible decisions through personalized carbon intelligence.

4. Product Mission

Reduce personal carbon emissions by helping users understand their environmental impact and guiding them toward practical, sustainable actions.

5. Challenge Alignment

The solution directly addresses:

Design a solution that helps individuals understand, track, and reduce their carbon footprint through simple actions and personalized insights.

Understand

Users gain visibility into:

- Carbon score
 - Emission categories
 - High-impact activities
 - Sustainability patterns
-

Track

Users monitor:

- Daily emissions
 - Monthly emissions
 - Historical trends
 - Sustainability goals
 - Recommendation adoption
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Reduce

Users receive:

- Personalized recommendations
- Impact forecasts
- Action plans
- Sustainability challenges

6. Target Audience

Primary Audience

Students

Characteristics:

- Technology friendly
- Sustainability curious
- Budget conscious

Goals:

- Learn sustainable habits
 - Reduce environmental impact
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Working Professionals

Characteristics:

- Limited time
- Data-driven

Goals:

- Efficient sustainability improvements
 - Automated tracking
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Families

Characteristics:

- Shared consumption patterns

Goals:

- Reduce household emissions
 - Build sustainable lifestyles
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7. User Personas

Persona 1

Name: Rahul

Age: 21

Occupation: Engineering Student

Pain Points:

- Doesn't understand carbon footprint
- Wants practical sustainability guidance

Goals:

- Lower transportation emissions
 - Learn sustainability fundamentals
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Persona 2

Name: Priya

Age: 29

Occupation: Software Engineer

Pain Points:

- Limited time
- Too much generic advice online

Goals:

- Receive personalized recommendations
 - Track measurable progress
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8. Product Objectives

Primary Objectives

- Improve carbon footprint awareness
 - Deliver personalized sustainability insights
 - Encourage behavior change
 - Track environmental progress
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Secondary Objectives

- Improve sustainability literacy
 - Increase user engagement
 - Build long-term habits
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9. Core Features

Carbon Intelligence Dashboard

Provides:

- Carbon Score
- Emission Breakdown
- Trend Analysis
- Sustainability Rating

Purpose:

Help users understand environmental impact.

Carbon Calculator

Categories:

Transportation

Examples:

- Cars
- Bikes

- Flights
 - Public Transport
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Food

Examples:

- Meat Consumption
 - Dairy Consumption
 - Vegetarian Meals
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Energy

Examples:

- Electricity Usage
 - Heating
 - Cooling
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Shopping

Examples:

- Clothing
 - Electronics
 - Household Goods
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Purpose:

Generate accurate carbon estimates.

Personalized Recommendation Engine

Purpose:

Generate recommendations based on individual behavior.

Examples:

Transportation Heavy User:

Recommendation:

- Bike twice weekly

Estimated Impact:

- 120 kg CO₂ reduction annually
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Food Heavy User:

Recommendation:

- Replace two meat meals weekly

Estimated Impact:

- 90 kg CO₂ reduction annually
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Sustainability Goals

Users can:

- Create goals
 - Track progress
 - Complete milestones
 - Monitor reductions
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Carbon Twin Simulator

Purpose:

Predict future emissions.

Examples:

"What happens if I switch to public transport?"

"What happens if I reduce meat consumption?"

Output:

- Predicted emissions

- Estimated savings
 - Sustainability forecast
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Sustainability Challenges

Examples:

- Car-Free Week
- Meat-Free Month
- Energy Saver Challenge

Purpose:

Increase engagement and behavior change.

10. Functional Requirements

FR-01

Users shall create accounts.

FR-02

Users shall calculate carbon emissions.

FR-03

Users shall view carbon breakdowns.

FR-04

Users shall receive personalized recommendations.

FR-05

Users shall track sustainability goals.

FR-06

Users shall view historical trends.

FR-07

Users shall simulate future sustainability scenarios.

FR-08

Users shall participate in sustainability challenges.

11. Non-Functional Requirements

Code Quality (High Impact)

Requirements:

- TypeScript throughout frontend
 - Modular architecture
 - SOLID principles
 - Clear folder structure
 - Reusable components
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Security (Medium Impact)

Requirements:

- Helmet security headers
 - JWT authentication
 - bcrypt password hashing
 - Input validation
 - Rate limiting
 - Secure environment variables
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Efficiency (Medium Impact)

Requirements:

- PostgreSQL indexing
 - Redis caching
 - Pagination
 - Lazy loading
 - Optimized API responses
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Testing (Low Impact)

Requirements:

Unit Testing

Tool:

Jest

Integration Testing

Tool:

Supertest

End-to-End Testing

Tool:

Playwright

Coverage:

- Registration
 - Login
 - Dashboard
 - Recommendations
 - Goal Tracking
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Accessibility (Low Impact)

Requirements:

- WCAG 2.2 AA
 - Semantic HTML
 - Keyboard Navigation
 - Screen Reader Support
 - High Contrast Design
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12. Success Metrics

Product Metrics

- Daily Active Users
 - Monthly Active Users
 - Retention Rate
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Sustainability Metrics

- Carbon Reduction Achieved
 - Recommendation Adoption Rate
 - Goal Completion Rate
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Technical Metrics

- API Response < 500ms
 - Dashboard Load < 2s
 - Uptime > 99.9%
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13. Future Roadmap

Phase 1

- Carbon Calculator
 - Dashboard
 - Authentication
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Phase 2

- Recommendation Engine
 - AI Carbon Coach
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Phase 3

- Goals
 - Challenges
 - Achievements
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Phase 4

- Carbon Twin
 - Forecasting Engine
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14. Assumptions

The following assumptions were made:

- Users will manually enter activity data initially
 - Carbon emission factors will come from publicly available datasets
 - Recommendations are advisory only
 - Carbon calculations are estimates, not regulatory measurements
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15. Definition of Success

CarbonWise is successful when users:

- Understand their carbon footprint
- Identify emission hotspots
- Receive personalized sustainability guidance
- Take measurable actions
- Reduce emissions over time
- Develop long-term sustainable habits